

MATLAB: Session 4

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Exercises

Part A – Linear Systems

A.1 Consider the parametric linear system, for $k \in \mathbb{R}$:

$$\begin{cases} x + y + z = 1, \\ x + ky + z = 2, \\ 2x + (k + 1)y + 3z = 3. \end{cases}$$

Check in MATLAB for which values of a the system is compatible. For those such values of k , solve the linear system.

Part B – Linear Applications

In the following, $\mathcal{E}_n = \{e_1, \dots, e_n\}$ denotes the canonical basis for $\mathbb{R}^n$

B.1 Consider the endomorphism of $\mathbb{R}^3$, $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, described as

$$\begin{cases} f(e_1) = 2e_1 + e_2 - 3e_3 \\ f(e_2) = e_1 + 3e_2 - 2e_3 \\ f(e_3) = f(e_1) - 2f(e_2) \end{cases}.$$

B.1.1 Assign to the variable `B` the matrix associated to f with respect to $\mathcal{E}_3$.

B.1.2 Compute the dimension of $\ker f$ and assign it to the variable `dimkerf`.

B.1.3 Compute the dimension of $\operatorname{Im} f$ and assign it to the variable `dimimf`.

B.1.4 Compute a basis for $\ker f$ and assign it to the variable `basiskerf`.

B.1.5 Compute a basis for $\operatorname{Im} f$ and assign it to the variable `basisimf`.

B.1.6 Is f injective? Is f surjective?

B.2 Consider the endomorphism of $\mathbb{R}^3$

$$g(x, y, z) = (x - z, 2x + y, 2y + x).$$

- B.2.1** Assign to the variable `C` the matrix associated to g with respect to $\mathcal{E}_3$.
- B.2.2** Compute the dimension of $\ker g$ and assign it to the variable `dimker $g$` .
- B.2.3** Compute the dimension of $\text{Im} g$ and assign it to the variable `dimim $g$` .
- B.2.4** Compute a basis for $\ker g$ and assign it to the variable `basisker $g$` .
- B.2.5** Compute a basis for $\text{Im} g$ and assign it to the variable `basisim $g$` .
- B.2.6** Is g injective? Is g surjective?

B.3 Consider the endomorphism of $\mathbb{R}^3$

$$h(x, y, z) = (2x, z - y, y + x).$$

- B.3.1** Assign to the variable `D` the matrix associated to h with respect to $\mathcal{E}_3$.
- B.3.2** Compute the dimension of $\ker h$ and assign it to the variable `dimker $h$` .
- B.3.3** Compute the dimension of $\text{Im} h$ and assign it to the variable `dimim $h$` .
- B.3.4** Compute a basis for $\ker h$ and assign it to the variable `basisker $h$` .
- B.3.5** Compute a basis for $\text{Im} h$ and assign it to the variable `basisim $h$` .
- B.3.6** Is h injective? Is h surjective?

B.4 Consider the endomorphism of $\mathbb{R}^3$, $s_m : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, described as

$$\begin{cases} s_m(e_1) = me_1 + e_2 \\ s_m(e_2) = 4e_1 + me_2 \\ s_m(e_3) = -e_1 - e_2 + 2e_3 \end{cases},$$

where m is a real parameter.

- B.4.1** Assign to the variable `E_m` the matrix associated to s_m with respect to $\mathcal{E}_3$.
- B.4.2** Compute the dimension of $\ker s_m$ for each m , and assign it to the variables `dimkers $_m$` .
- B.4.3** Compute the dimension of $\text{Im} s_m$ and assign it to the variables `dimims $_m$` .
- B.4.4** Select an m for which s_m is not surjective. Substitute this m to `E_m`, defining a new variable `E`.
- B.4.5** For such an m , compute a basis for $\ker s_m$ and a basis for $\text{Im} s_m$ and assign them to the variables `basiskers` and `basisims`, respectively.